

NEGATIVE VOLTAGE GENERATOR

Analog Electronics Lab Project

Using an Oscillator and a Diode Pump

Objective

To design and validate a circuit that derives a usable negative supply rail from a single positive DC source, using only a 555 timer oscillator and a discrete diode-pump (charge-pump) stage — without a bipolar power supply or an inverting DC-DC converter IC.

Circuit and Simulation

Fig. 1 – Diode-pump negative voltage generator, combining a 555 timer astable oscillator (R1, R2, C3, C4) with a two-diode, two-capacitor charge pump (D1, D2, C1, C2).

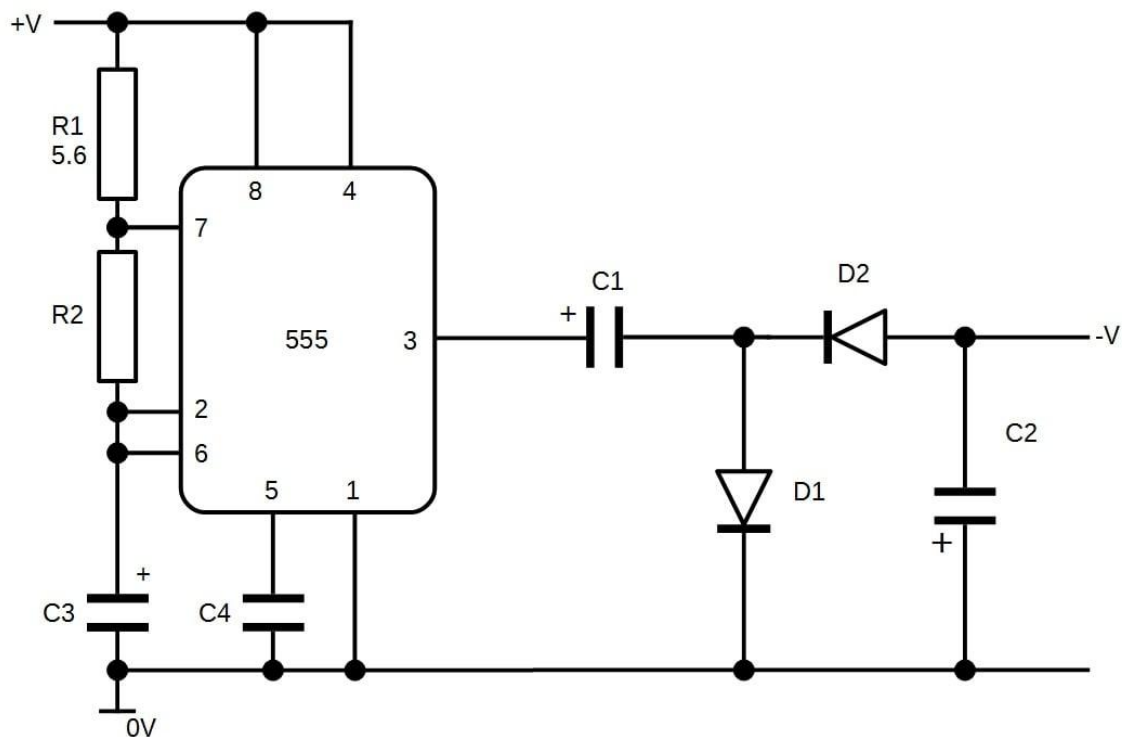


Fig. 2 – LTspice simulation schematic used to verify the circuit before hardware implementation.

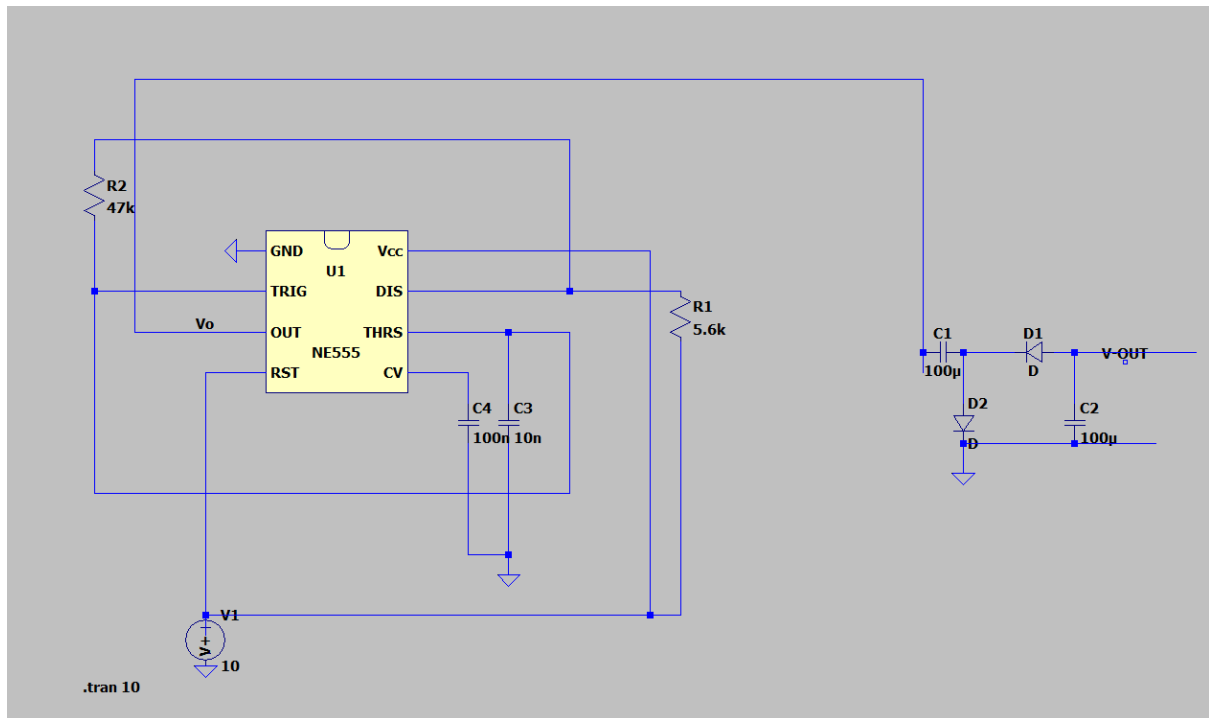
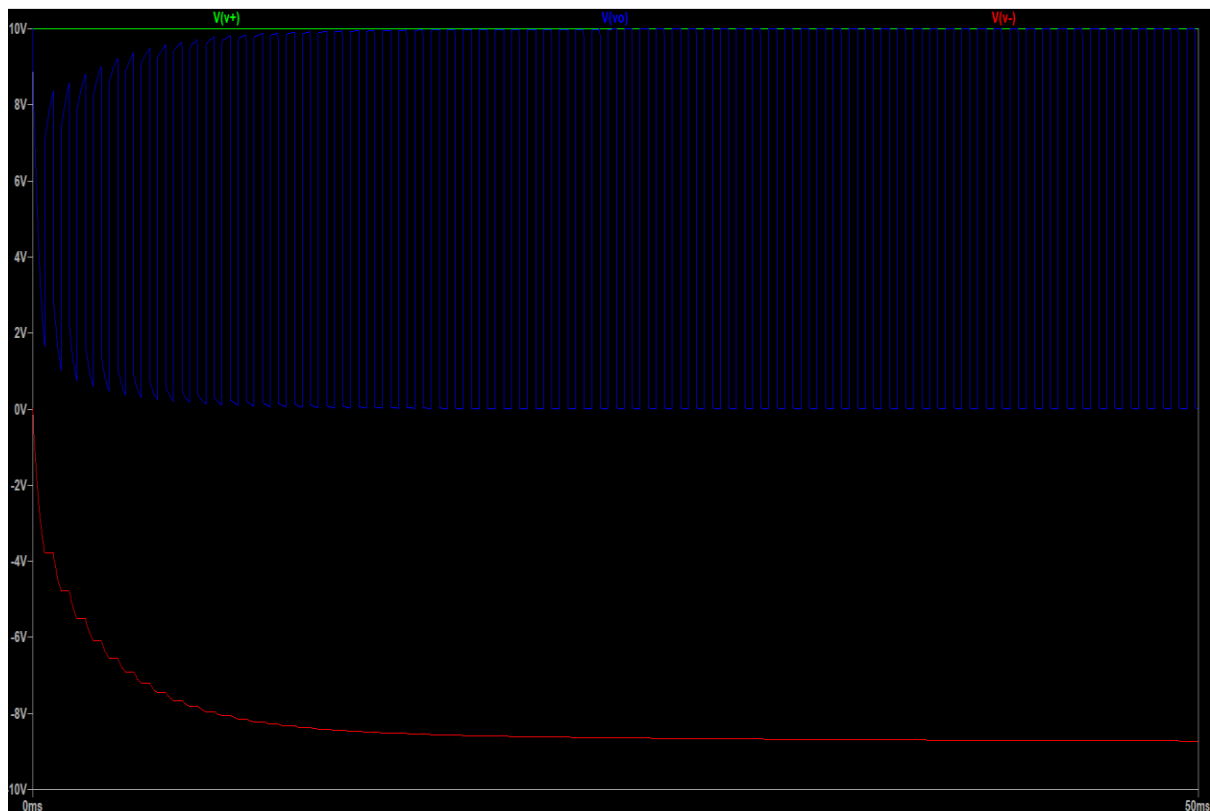


Fig. 3 – Transient simulation: V(v+) is the steady 10 V supply, V(vo) is the oscillator's square-wave output, and V(v-) is the negative output settling to about -8.7 V.



Bill of Materials – Diode Pump Circuit

Component	Identifier	Quantity
Diode (Generic Silicon)	D1, D2	2
10uF Electrolytic Capacitor	C1	1
100uF Electrolytic Capacitor	C2	1

Bill of Materials – 555 Oscillator Stage

Component	Identifier	Quantity
555 Timer IC	555	1
5.6kΩ Resistor	R1	1
47kΩ Resistor	R2	1

100nF Capacitor	C4	1
10nF Capacitor	C3	1
Assorted Wires	-	-
Breadboard	-	1

Working Principle

The negative voltage generator uses the concept of capacitor coupling to produce a negative voltage from a single DC source. Capacitor coupling is the phenomenon in which a capacitor tries to maintain the voltage difference across it at all costs.

We utilise a couple of diodes and capacitors in the diode-pump circuit to conserve this voltage difference and create a negative potential at the output. However, this negative voltage gets used up almost instantaneously. Thus we require a continuous source of fluctuating voltage (preferably a square wave) at the input.

To achieve this we construct our own oscillator using a 555 timer, a few resistors and capacitors. This oscillator takes in a DC input and produces a square wave that oscillates between 0 V and the input value. The output becomes more consistent over time, as seen in the graph. Thus we get a negative voltage at the output which is roughly equal in magnitude to the negative of the input voltage at the oscillator end.

Results and Observations

The transient simulation confirms proper operation of the circuit: the 555 oscillator produces a clean square wave ($V(v_o)$) swinging between 0 V and the 10 V supply rail, and the diode-pump output ($V(v_-)$) settles to approximately -8.7 V after a short charging transient of a few oscillator cycles. The roughly 1.3 V shortfall from the ideal -10 V is attributable to the forward-voltage drops across diodes D1 and D2 in the pump path.

Key Design Equation

$$F = 1.44 / [(R1 + 2R2)(C1 + C2)]$$

Applications

This oscillator-driven diode-pump technique is a compact way to generate an auxiliary negative rail for circuits that need a small bipolar supply without a dedicated dual-output power source — for example, biasing the negative rail of op-amp based analog front-ends, supplying RS-232 line drivers, or generating a small negative bias for sensor or LCD circuits.